

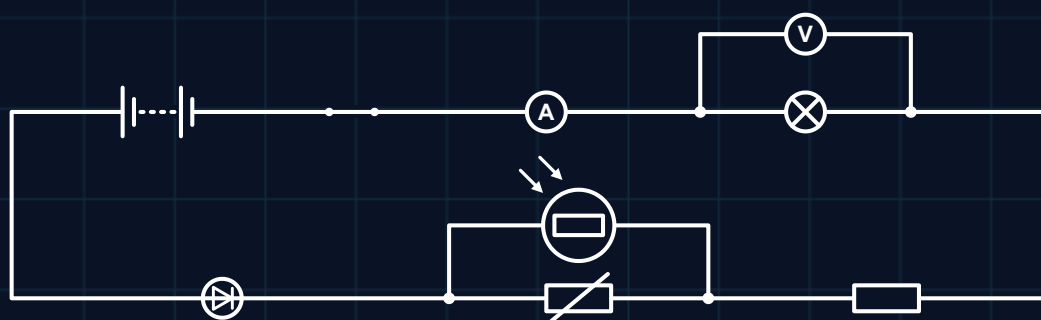
# ELECTRIC CIRCUITS

## LESSON 11 POWER AND ENERGY TRANSFER

Which transfers more energy: a 3 kW kettle boiling for 3 minutes, or a 10 W LED lamp left on for 5 hours?

### IN THIS LESSON YOU WILL

- describe power as the energy transferred per second
- use  $E = P t$ ,  $P = V I$  and  $P = I^2 R$  with the right units
- describe the energy transfers in everyday appliances
- choose the right equation for a question



NAME

\_\_\_\_\_

CLASS

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TEACHER

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TARGET GRADE

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# HOW THIS BOOKLET WORKS

This is the booklet for **Lesson 11** of the Electric Circuits unit. It builds on the lessons before it; the “Do now” questions and the quick answers at the back help you check what you remember. Keep it: it is your classwork book *and* your revision guide.



## DO NOW

Quick questions from earlier lessons. Remembering again is what makes it stick.



## LEARN IT

The explanation, diagrams and worked examples. Read this again when you revise.



## TRY IT

Practice that builds up: first with support, then on your own.



## LAB

Record measurements from a virtual lab or a required practical.



## EXAM CORNER

An AQA-style question with a tip on how the marks are given.



## REVISE IT

Summary, cover-and-answer questions and an “I can” check to colour in.



**Revising at home?** (1) Read the Learn it pages. (2) Cover them and answer the “Cover the page” questions in Revise it. (3) Check with the quick answers at the back. (4) Do the matching mission on the companion website, which has more questions with answers.

## THE ROUTE THROUGH THE UNIT · THIS LESSON IS HIGHLIGHTED

**01** Complete circuits and symbols

**03** Potential difference

**05** Resistance of a wire RP

**07** Resistors in series and parallel RP

**09** Investigating I–V characteristics RP

### **11** Power and energy transfer

Power,  $E = P t$ ,  $P = V I$ ,  $P = I^2 R$ , energy transfers in appliances · Mission 7 · Power & the National Grid

**02** Current and charge

**04** Resistance and  $V = I R$

**06** Series and parallel circuits

**08** I–V graphs

**10** Thermistors, LDRs and sensors

**12** Mains, safety and the National Grid

Companion website: [panphy.app/gcsephy/year10phy/unit01](https://panphy.app/gcsephy/year10phy/unit01) · missions, interactive equation triangles, required-practical pages and the Exam Zone.

# POWER AND ENERGY TRANSFER

AQA Combined 6.2.4.1–6.2.4.2 · Physics 4.2.4.1–4.2.4.2

**BIG QUESTION** Which transfers more energy: a 3 kW kettle boiling for 3 minutes, or a 10 W LED lamp left on for 5 hours?

## BY THE END YOU CAN

- describe power as the energy transferred per second
- use  $E = P t$ ,  $P = V I$  and  $P = I^2 R$  with the right units
- describe the energy transfers in everyday appliances
- choose the right equation for a question

power

watt (W)

kilowatt (kW)

energy transferred

power rating

## ONLINE COMPANION

### Mission 7 · Power & the National Grid

Read the power and energy sections, then try questions 1–4.

[panphy.app/gcsephy/year10phy/unit01/lesson/power-and-national-grid/](https://panphy.app/gcsephy/year10phy/unit01/lesson/power-and-national-grid/)



RETRIEVE

**DO NOW**

From earlier lessons. No notes!

1

5.0 C passes through a 12 V lamp. How much energy is transferred?


2

2.0 A flows for 1 minute.  
How much charge flows?


3

What is the resistance of a heater with 230 V across it and 5.0 A through it?



EXPLAIN

## HOW FAST IS ENERGY TRANSFERRED?

**Power** is the **energy transferred per second**. It is measured in **watts (W)**: 1 W = 1 J per second. A 2000 W (2 kW) kettle transfers 2000 J of energy every second.

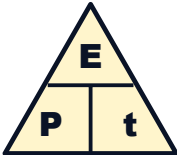
| APPLIANCE       | TYPICAL POWER   | ENERGY TRANSFERRED EACH SECOND |
|-----------------|-----------------|--------------------------------|
| LED lamp        | 10 W            | 10 J                           |
| Laptop          | 60 W            | 60 J                           |
| Hairdryer       | 1.2 kW = 1200 W | 1200 J                         |
| Kettle          | 2.2 kW = 2200 W | 2200 J                         |
| Electric shower | 9 kW = 9000 W   | 9000 J                         |

Every appliance transfers energy electrically from the mains to other energy stores: a kettle to the **thermal store** of the water; a hairdryer's fan to **kinetic** energy and its heater to thermal; a phone charger to the **chemical store** of the battery.

**energy transferred = power × time**

**$E = P t$**

E in joules (J) · P in watts (W) · t in seconds (s)  
Convert kW → W (× 1000) and minutes or hours → seconds first.

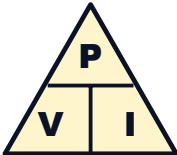


|                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>WORKED EXAMPLE</b></p> <p><b>A 2.0 kW kettle is on for 3.0 minutes. Calculate the energy transferred.</b></p> <p>KNOW      P = 2000 W, t = 180 s</p> <hr/> <p>EQUATION    <math>E = P t</math></p> <hr/> <p>SUBSTITUTE <math>E = 2000 \times 180</math></p> <hr/> <p>ANSWER      <b><math>E = 360\,000\text{ J}</math></b></p> | <p><b>YOUR TURN</b></p> <p><b>A 1.2 kW hairdryer is used for 10 minutes. Calculate the energy transferred.</b></p> <p>KNOW</p> <hr/> <p>EQUATION</p> <hr/> <p>SUBSTITUTE</p> <hr/> <p>ANSWER</p> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**power = p.d. × current**

**$P = V I$**

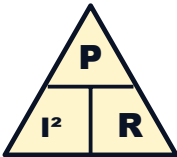
P in watts (W) · V in volts (V) · I in amperes (A)



**power = current<sup>2</sup> × resistance**

**$P = I^2 R$**

P in watts (W) · I in amperes (A) · R in ohms (Ω)



**WORKED EXAMPLE**

A heater on the 230 V mains draws a current of 4.0 A. Calculate its power.

EQUATION  $P = VI$

SUBSTITUTE  $P = 230 \times 4.0$

ANSWER  **$P = 920 \text{ W}$**

**YOUR TURN**

A current of 3.0 A flows through a  $20 \Omega$  resistor. Calculate the power.

EQUATION

SUBSTITUTE

ANSWER

**!** In  $P = I^2 R$ , **square the current first**. Doubling the current makes the heating **four times** bigger: that is why cables carrying big currents get hot.

| IF YOU KNOW...         | USE         |
|------------------------|-------------|
| p.d. and current       | $P = VI$    |
| current and resistance | $P = I^2 R$ |
| power and time         | $E = Pt$    |
| charge and p.d.        | $E = QV$    |
| current and time       | $Q = It$    |



PRACTISE

**TRY IT**

**1** Answer the big question. Show both calculations.

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2

**SUPPORTED** A 12 V car headlamp draws a current of 5.0 A. Calculate its power.

|            |       |
|------------|-------|
| EQUATION   |       |
| SUBSTITUTE |       |
| ANSWER     | unit: |

3

**SUPPORTED** A 9.0 kW electric shower runs for 8.0 minutes. Calculate the energy transferred in joules.

|            |       |
|------------|-------|
| CONVERT    |       |
| EQUATION   |       |
| SUBSTITUTE |       |
| ANSWER     | unit: |

4

**ON YOUR OWN** A current of 2.0 A flows through a 10  $\Omega$  resistor. Then the current is increased to 4.0 A. Calculate the power each time. What happens to the power when the current doubles?

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5

**ON YOUR OWN** A device transfers 240 J of energy when 40 C of charge flows through it. Calculate the p.d. across it.

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6

**CHALLENGE** A lamp is rated 60 W, 230 V. Calculate the current through it and its resistance.

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**Get the marks:** Write the power in watts before you substitute:  $2.2 \text{ kW} = 2200 \text{ W}$ . Give answers to a sensible number of significant figures.

7

A 2.2 kW kettle runs on the 230 V mains. (a) Calculate the current [6 marks]  
through the kettle. (b) Calculate the energy transferred in 150 s. (c) A heater draws  
twice this current through the same resistance. How does its heating power  
compare?

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**SUMMARY**

- Power is energy transferred per second.  $1 \text{ W} = 1 \text{ J/s}$ .
- $E = P t$  ( $P$  in W,  $t$  in s).
- $P = V I$  and  $P = I^2 R$ .
- Doubling the current through a resistor makes the heating 4 times bigger.
- Appliances transfer energy electrically to thermal, kinetic or chemical stores.

**COVER THE PAGE AND ANSWER****1** What is power?

.....  
**3** Write two equations for electrical power.  
.....

**2** Convert 3.5 kW into watts.

.....  
**4** Write the equation linking energy, power and time.  
.....

**I CAN...**

describe power as energy transferred per second

use  $E = P t$  with unit conversionsuse  $P = V I$  and  $P = I^2 R$ 

choose the right equation from the data given



# EQUATION TOOLKIT

The equations you have met so far. You will meet these one at a time. Learn each equation **in words** as well as symbols: your exams may or may not give you an equation sheet, and knowing them by heart makes every question faster.

$$Q = I t$$

Lesson 2

charge flow = current × time

C, A, s

Rearranged:  $I = Q \div t$  ·  $t = Q \div I$

$$E = Q V$$

Lesson 3

energy transferred = charge ×  
p.d.

J, C, V

Rearranged:  $V = E \div Q$  ·  $Q = E \div V$

$$V = I R$$

Lesson 4

p.d. = current × resistance

V, A,  $\Omega$

Rearranged:  $I = V \div R$  ·  $R = V \div I$

$$R_{\text{total}} = R_1 + R_2$$

Lesson 6

total resistance in series

$\Omega$

Rearranged: (series only)

$$P = V I$$

Lesson 11

power = p.d. × current

W, V, A

Rearranged:  $V = P \div I$  ·  $I = P \div V$

$$P = I^2 R$$

Lesson 11

power = current<sup>2</sup> × resistance

W, A,  $\Omega$

Rearranged:  $R = P \div I^2$  ·  $I = \sqrt{P \div R}$

$$E = P t$$

Lesson 11

energy transferred = power ×  
time

J, W, s

Rearranged:  $P = E \div t$  ·  $t = E \div P$

## FOUR STEPS FOR EVERY CALCULATION

1

Write down what you know, with units. Convert units first (minutes → seconds, kW → W, mA → A).

2

Write the equation you will use, in symbols.

3

Substitute the numbers, then rearrange if needed.

4

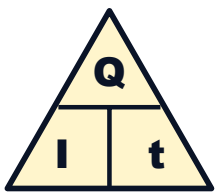
Give the answer with a unit. Ask: is it sensible?

| CONVERT                                              | HOW    | EXAMPLE            |
|------------------------------------------------------|--------|--------------------|
| minutes → seconds                                    | × 60   | 3.0 min = 180 s    |
| hours → seconds                                      | × 3600 | 2.0 h = 7200 s     |
| milliamps (mA) → amps (A)                            | ÷ 1000 | 250 mA = 0.25 A    |
| kilowatts (kW) → watts (W) · kilohms (kΩ) → ohms (Ω) | × 1000 | 2.2 kW = 2200 W    |
| kilovolts (kV) → volts (V)                           | × 1000 | 400 kV = 400 000 V |

## EQUATION TRIANGLES: HELP WITH REARRANGING

- 1 Cover the quantity you want to find.
- 2 The two left **side by side**? **Multiply** them.
- 3 One **above** the other? **Divide** the top by the bottom.

Example: to find  $I$  from  $Q = I t$ , cover  $I$ .  $Q$  is above  $t$ , so  $I = Q \div t$ . Triangles only help with rearranging: you still need to learn each equation. For  $P = I^2 R$ , the triangle gives  $I^2$ ; take the square root to find  $I$ .

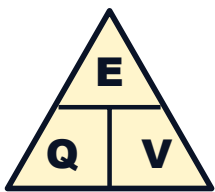


$$Q = I t$$

$$Q = I \times t$$

$$I = Q \div t$$

$$t = Q \div I$$

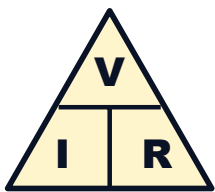


$$E = Q V$$

$$E = Q \times V$$

$$Q = E \div V$$

$$V = E \div Q$$

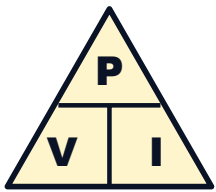


$$V = I R$$

$$V = I \times R$$

$$I = V \div R$$

$$R = V \div I$$

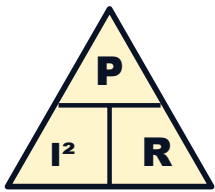


$$P = V I$$

$$P = V \times I$$

$$V = P \div I$$

$$I = P \div V$$



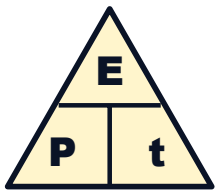
$$P = I^2 R$$

$$P = I^2 \times R$$

$$R = P \div I^2$$

$$I^2 = P \div R,$$

$$\text{so } I = \sqrt{(P \div R)}$$



$$E = P t$$

$$E = P \times t$$

$$P = E \div t$$

$$t = E \div P$$

Interactive version: [panphy.app/gcsephy/year10phy/unit01/equation-triangles](https://panphy.app/gcsephy/year10phy/unit01/equation-triangles).

## REFERENCE

# KEY WORDS FROM THIS LESSON

Learn the meaning and spelling. In exams, precise words earn marks: say “p.d. across” and “current through”.

**Power (P)** · Energy transferred per second. Unit: watt (W).

# QUICK ANSWERS

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in your teacher's answer edition.

**DO NOW**

1.  $E = Q V = 60 \text{ J}$
2.  $Q = 2.0 \times 60 = 120 \text{ C}$
3.  $R = 230 \div 5.0 = 46 \Omega$

**COVER THE PAGE**

1. Energy transferred per second.
2.  $3500 \text{ W}$
3.  $P = V I$  and  $P = I^2 R$
4.  $E = P t$